Commensurability of Observation Protocols: When Coherent Windows Determine a Process

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Abstract

Observing a process through overlapping finite windows yields local statistics; we ask when locally coherent window-statistics must arise from a single global process. An observation protocol is *commensurable* when empirical adequacy (pairwise coherence, EA) forces realisability by a declared class (PR($\mathcal{R}$)); the failure is contextuality in the sense of Abramsky–Brandenburger, taken from the observational side. Geometrically, a protocol is commensurable iff its coherence polytope C equals its realisability polytope R, so contextuality is an exactly computable separation with a rational witness. For protocols whose contexts are windows on a symbolic dynamical system we prove: a *trichotomy* (acyclic reporting always commensurable; full-support coordinate protocols commensurable; variety-typed protocols realise every rational polytope, hence admit no finite classification, by De Loera–Onn); a *parity theorem* (the golden-mean shift on a ring of length L is commensurable iff L is even, the odd-ring gap a single extreme point, its own PR-box); ten *taming* mechanisms; and, assembled from these, *Circuit Localization* for the symmetric loopless marginal-determined languages, where contextuality is excluded by structure rather than search (the loopy and general-cyclic extensions verified only to finite scale). A winding criterion characterises the unsafe lengths as those carrying a layer-injective winding-≥ 2 cycle; its combinatorial kernel is machine-verified in Lean 4. The open case — a universal impossibility conjecture for the strongly-connected non-symmetric class — is stated with its reduction to a pruning lemma and the gap that reduction carries.

1 Introduction

A process in time is often accessible only through *windows*: overlapping finite views, each recording the joint behaviour of finitely many coordinates over finitely many steps. The resulting local probability tables may agree wherever they overlap yet be generated by no single underlying process — the observational face of contextuality [2]. We ask, for symbolic dynamical observation, when local coherence forces global realisability, and answer it for a large structured class.

A finite alphabet A and finite supports $F \subseteq \mathbb{Z}$ give *contexts*, each a partition of A^F; a *protocol* is a family of contexts, and window-data assigns probabilities to the cells of each context.

Definition 1.1 (EA, PR($\mathcal{R}$), commensurability)**.** Window-data is *empirically adequate* (EA) if it is pairwise coherent: the contexts agree on the common subalgebra of each pair. It is *probabilistically realisable* in a class $\mathcal{R}$ (PR($\mathcal{R}$)) if some member of $\mathcal{R}$ (all measures / stationary / supported on a subshift / memory-m Markov, an observer's commitment lattice) induces it. A *model* is *contextual* when it is EA but not PR($\mathcal{R}$). A protocol is *commensurable* when EA $\Rightarrow$ PR($\mathcal{R}$) for all coherent data.

The two protocol axes are *reporting* (window layout) and *coarsening* (compression); the commitment class $\mathcal{R}$ is the third, graded, axis.

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On a finite outcome set the coherent data form a polytope C (the local/marginal polytope [18]) and the realisable data the polytope $R = \mathrm{conv}\{v(x)\}$, the convex hull of the cell-incidence vectors of the global configurations, with $R \subseteq C$.

Proposition 1.2 (commensurability is $C = R$)**.** *A protocol is commensurable iff $C = R$. A contextual model is a point of $C \setminus R$; over rational data any separating facet certifies non-realisability.*

Both polytopes are enumerable in exact arithmetic, so a commensurability verdict is a proof that $C = R$ or an exhibited rational witness in $C \setminus R$ (§3.1).

The polyhedral facts are classical: C and R are the marginal polytopes of Abramsky–Brandenburger's presheaf model [2]; acyclic extendability is Vorob'ev [17]; the constrained case is De Loera–Onn [8] and Průša–Werner [15]; the parity mechanism is stable-set integrality [13, 7, 11]. Over this sits an observational and dynamical layer: a cyclic protocol read as recurrence, so that a length is contextual exactly when the layered ring carries a winding-≥ 2 orbit (§2.4), and the tamings assembled into a localisation of the polytope (§2.3). The winding criterion and the localisation assembly sit above the satisfiability-width line of Barto–Kozik [5] and the tightness criterion of Thapper–Živný [16]; the signable tamings are the signed-minor tightness conditions of Weller [19, 20] read through the frame's bipartition.

2 The structure theory

2.1 The trichotomy

Write a symbolic language as a relation ρ (the legal adjacent pairs of a shift of finite type); a protocol reports windows on the configurations legal for ρ. The general problem splits into three regimes.

Theorem 2.1 (acyclic reporting is tame; Vorob'ev)**.** *If the reporting hypergraph of a protocol is acyclic, the protocol is commensurable for every language ρ.*

This is Vorob'ev [17] in the present language: consistent marginals over a regular (acyclic) complex extend, and acyclicity is the running-intersection condition of database theory [6], the two identified via [1]. Conditional-product glue over a junction tree assembles the window-marginals into a global distribution in the variety of ρ, so frustration requires a cycle.

Theorem 2.2 (full-support coordinate protocols are tame)**.** *A k-context protocol of coordinate contexts with full product support is commensurable: the product measure realises any coherent margins.*

The third regime is wild: a protocol may carry a *constraint variety* — the configurations restricted by ρ — and there the local polytope inherits arbitrary structure.

Theorem 2.3 (variety-typed protocols are universal)**.** *For protocols with three or more coordinate contexts subject to a constraint variety, the achievable coherence polytopes realise every rational polytope up to affine equivalence. Consequently: no finite facet taxonomy exists, model membership is NP-complete already at three contexts, and no bounded certificate normal form can classify contextual models in general.*

Proof. The covariance map [4] makes the polytopes cut/correlation polytopes; by De Loera–Onn [8] every rational polytope is a face of a three-way transportation polytope, realised as imposed zeros, i.e. as constraint varieties; the NP-completeness floor is [14]. □

This is why the classification is posed at the protocol level (a $\forall\exists$ question over structured families) and not the model level, where it is intractable. The rest of the paper classifies protocol families where structure defeats the universality.

2.2 The parity theorem

The cleanest cyclic case is the golden-mean shift: ρ_{gm} forbids adjacent 1s. Observe it around a ring of length L (scope contexts = the L edge-windows of the cycle C_L).

Theorem 2.4 (parity)**.** *The golden-mean ring protocol on C_L is commensurable iff L is even.*

Proof. Reduction (marginal-determination on the golden-mean ring). Write $u_i \in [0,1]$ for the marginal $\Pr[x_i = 1]$ at ring site $i \in \mathbb{Z}_L$. On the variety $x_i = 1 \Rightarrow x_{i+1} = 0$, so the joint distribution of the adjacent pair (x_i, x_{i+1}) has cells $(0,0), (0,1), (1,0)$ with probabilities p_{00}, p_{01}, p_{10} and $p_{11} = 0$. The endpoint marginals give $p_{10} + p_{11} = u_i$ and $p_{01} + p_{11} = u_{i+1}$, whence, using $p_{11} = 0$ and $\sum p = 1$,

$$p_{10} = u_i, \qquad p_{01} = u_{i+1}, \qquad p_{00} = 1 - u_i - u_{i+1}.$$

The edge distribution is thus *affine* in (u_i, u_{i+1}) — the language is marginal-determined — and nonnegativity of p_{00} is exactly the edge constraint $u_i + u_{i+1} \le 1$. Adjacent-context coherence is the shared marginal u_{i+1}, automatic in these coordinates. Hence

$$C \;\cong\; \mathrm{FSTAB}(C_L) = \{u \in [0,1]^L : u_i + u_{i+1} \le 1 \ \forall i \in \mathbb{Z}_L\},$$

the fractional stable-set polytope of the cycle C_L, and $R \cong \mathrm{STAB}(C_L) = \mathrm{conv}\{\mathbf{1}_S : S \subseteq \mathbb{Z}_L \text{ independent}\}$, since the realisable $\{0,1\}$-marginal profiles are exactly the indicators of independent sets (a homomorphism $C_L \to \rho_{\mathrm{gm}}$ is a 1-labelling with no two adjacent, i.e. an independent set).

Parity of the incidence determinant. The edge constraints $u_i + u_{i+1} \le 1$ have coefficient matrix M, the $L \times L$ vertex–edge incidence matrix of C_L read cyclically: $M = I + P$ where P is the cyclic shift $Pe_i = e_{i+1}$. Then $\det M = \det(I + P) = \prod_{k=0}^{L-1}(1 + \omega^k)$ over the L-th roots of unity ω^k (P has eigenvalues the roots of unity), and $\prod_k (1 + \omega^k) = 1 - (-1)^L$ since $\prod_k (z - \omega^k) = z^L - 1$ evaluated at $z = -1$ gives $(-1)^L - 1$, so $\prod_k(-1 - \omega^k) = (-1)^L - 1$ and $\prod_k(1 + \omega^k) = (-1)^L\big((-1)^L - 1\big) = 1 - (-1)^L$. Thus

$$\det M \;=\; 1 - (-1)^L \;=\; \begin{cases} 0, & L \text{ even}, \\ 2, & L \text{ odd}. \end{cases}$$

Every *proper* square submatrix of M omits at least one ring edge, hence is the incidence matrix of a disjoint union of *paths* (a forest); a path incidence matrix is triangularisable by ordering vertices along the path, so its determinant lies in $\{-1, 0, 1\}$ — every proper square submatrix is totally unimodular.

Even case. For L even, C_L is bipartite, so M itself is totally unimodular (bipartite incidence matrices are TU, König/Hoffman–Kruskal). With integral right-hand sides $\mathrm{FSTAB}(C_L)$ has only integral vertices, and its integral points are exactly the independent-set indicators, so $\mathrm{FSTAB}(C_L) = \mathrm{STAB}(C_L)$: $C = R$, commensurable.

Odd case — the unique fractional vertex. For L odd, let u^* be a fractional (non-integral) vertex of $\mathrm{FSTAB}(C_L)$. A vertex is determined by L tight constraints among $\{u_i \ge 0\}$ and $\{u_i + u_{i+1} \le 1\}$. Suppose some coordinate is tight at $u_j^* = 0$. Deleting site j breaks the cycle: the remaining active constraints are those of a *path* on the other sites, whose incidence matrix is TU (forest), so every vertex of that face is integral — u^* would be integral, contradiction. Hence $u^* > 0$ coordinatewise, so no $u_i \ge 0$ constraint is tight, and all L tight constraints must be the edge constraints $u_i + u_{i+1} = 1$ for every $i \in \mathbb{Z}_L$. This linear system is $Mu = \mathbf{1}$; since L is odd $\det M = 2 \neq 0$, so it has the unique solution. Summing $u_i + u_{i+1} = 1$ around the ring gives $2\sum_i u_i = L$, and the cyclic alternation $u_{i+1} = 1 - u_i$ around an odd cycle forces $u_i = 1 - u_i$, i.e.

$$u^* \equiv \tfrac{1}{2}.$$

Thus $\mathrm{FSTAB}(C_L^{\mathrm{odd}})$ is $\mathrm{STAB}(C_L)$ with exactly one extra vertex $u \equiv \frac{1}{2}$: the EA/PR gap on an odd golden-mean ring is a single point. □ □

Corollary 2.5 (the odd-ring PR-box)**.** *For L odd the half-model $u \equiv \frac{1}{2}$ is separated from $R = \mathrm{STAB}(C_L)$ by the odd-cycle inequality $\sum_i u_i \le (L-1)/2$, with relative violation $1/(L-1)$; and it is strongly contextual — its contextual fraction is* 1*. It is the golden-mean ring's analogue of a PR-box.*

Proof. Separating facet. Every independent set S of the odd cycle C_L has $|S| \le (L-1)/2$ (an independent set misses at least one vertex of each of the $\lceil L/2 \rceil$ disjoint-as-possible edges; the maximum independent set of an odd cycle has size $(L-1)/2$), so $\sum_i u_i \le (L-1)/2$ holds at every vertex of R, hence on all of R. At $u \equiv \frac{1}{2}$ the left side is $L/2$, exceeding $(L-1)/2$ by $\frac{1}{2}$; relative to the bound this is $\frac{1}{2}/\frac{L-1}{2} = 1/(L-1)$.

Contextual fraction 1. The contextual fraction is $1 - \lambda^*$, where $\lambda^* = \max\{\lambda : \ u - \lambda\nu \ge 0$ cellwise for some $\nu \in R\}$ is the largest realisable part fitting under u. On each edge u places mass $\frac{1}{2}$ on $(1,0)$, $\frac{1}{2}$ on $(0,1)$, and 0 on $(0,0)$. Every $\nu \in R$ is a mixture of independent-set configurations, and any configuration omitting both endpoints of some edge puts mass on that edge's $(0,0)$ cell, which u forbids. So an admissible ν must occupy exactly one endpoint of every edge: $v_i + v_{i+1} = 1$ for all $i \in \mathbb{Z}_L$. Summing gives $2\sum_i v_i = L$, impossible for integer $\sum_i v_i$ when L is odd. Hence $\lambda^* = 0$ and u is strongly contextual. □ □

Remark 2.6 (dynamical reading)**.** Sliding edge-windows on a period-L orbit identify time modulo L, so a cyclic scope structure is recurrence and Theorem 2.4 says golden-mean dynamics produces contextual data exactly on odd-period recurrence. The separating facet is the odd-cycle inequality, one instance of the n-cycle noncontextuality family [3] whose $n = 5$ case is the KCBS bound [12]; the parity split is the golden-mean face of that dichotomy, reached by recurrence rather than exclusivity.

The spectrum varies with the language: $\{\le\}$ is tame on all rings; XOR on even rings (odd rings have empty variety); the full language on none. More constraint can mean more tame.

2.3 Tamings and Circuit Localization

A commensurable language keeps the winding obstruction of §2.4 from forming — FSTAB = STAB with no fractional vertex. It can do so in one of two ways: by *exhibiting* integrality directly (a terminal certificate), or by *stripping* structure until a terminal certificate applies (a reduction). The mechanisms we catalogue fall under this dichotomy; each is a route to the same one-point criterion, not an independent phenomenon.

Definition 2.7 (the taming mechanisms)**.** A protocol/language is tamed by one of the following. *Terminal integrality certificates:*

(1) *chains / empty-intersection*: the pointwise bound $\sum_t \mathbf{1}_{c_t} \le m-1$ (§2.2 base case);

(2) *cross-context cliques*: perfect-graph coverage integrality;

(3) *signability*: total unimodularity from a $\{\pm 1\}$ column signing along the frame's bipartition, by Hoffman–Gale (appendix to [10]; cf. Ghouila-Houri [9]), with König as the edge case;

(4) *twin expansion*: a below-total-unimodularity integrality mechanism (a phase-forced decomposition), realised by a constructive θ-sweep.

Reductions to a terminal certificate:

(5) *transient branching*: branching off the recurrent core is deleted;

(6) *direct-sum decomposition*: a disconnected relation splits componentwise;

(7) *grading*: the transfer digraph fibered over $\mathbb{Z}_d$ — equivalently, Perron–Frobenius imprimitivity of period d;

(8) *monotone collapse*: a non-symmetric relation induces monotone potentials that strong connectivity flattens (the SCC-reduction lemma);

(9) *bipartite-target phase decoupling*: a bipartite target is a fibered join of two copies of its edge polytope over a global phase, reducing $C = R$ to the signable certificate (3).

The remaining case, *determinism* (a functional transfer relation), is degenerate: the polytope collapses to a point.

The dichotomy is not merely presentational for two of the entries: grading is *provably* imprimitivity, and phase decoupling *provably* reduces to signability (Lemma 2.10). The others are grouped by the $C = R$ mechanism they exhibit or invoke, not by their realisation — three of them share the θ-sweep as a construction while resting on different integrality grounds. The signable certificate is representative.

Theorem 2.8 (signable languages are tame on bipartite frames)**.** *Let a language be* marginal-determined *(its cell simplex is affine in the endpoint marginals) and* signable *(the edge-constraint matrix admits a $\{\pm 1\}$ column signing along the frame's bipartition). Then on every bipartite frame it is commensurable. The realisation is constructive: an anti-comonotone θ-sweep (threshold θ on one bipartition class, $1-\theta$ on the other) produces an exact mixture of global configurations reproducing any coherent point.*

Proof. There are two claims: $C = R$, and a constructive realisation.

$C = R$ (integrality). Fix a bipartite frame with 2-colouring $V = V_+ \sqcup V_-$. Marginal-determination lets us use vertex-marginal coordinates; the coherence polytope C is then cut out by the edge-nonnegativity inequalities of the language together with the marginal-normalisation equalities. Signability is the hypothesis that the coefficient matrix M of the edge-constraint rows admits a $\{\pm 1\}$ column signing: multiplying the columns indexed by V_- by -1 turns M into a matrix each of whose rows has its nonzero entries of *opposite* sign, because every edge of the frame joins a V_+ vertex to a V_- vertex (this is exactly where 2-colourability enters). A $\{0, \pm 1\}$ matrix in which every column can be signed so that every row's two nonzeros are of opposite sign is totally unimodular — this is the sufficient condition of Hoffman and Gale (appendix to [10]; equivalently Ghouila-Houri's interval-signing criterion [9]). With integral right-hand sides (the marginal constraints have $\{0, 1\}$ data), a totally unimodular system has only integral vertices. Each integral vertex of C is a $\{0, 1\}$ vertex-marginal profile, i.e. a global configuration, so it lies in $R = \mathrm{conv}\{v(x)\}$; hence $C = R$.

The θ-sweep realisation. Integrality gives $C = R$ non-constructively; the sweep exhibits, for any coherent point $q \in C$, an explicit probability mixture of global configurations inducing it. Parametrise by a threshold $\theta \in [0, 1]$. On the class V_+ read a state off q's marginals by the *comonotone* rule — accumulate the marginal mass in a fixed state order and select the state whose cumulative interval contains θ; on V_- read by the *anti-comonotone* rule with threshold $1-\theta$ (the reversed order). For each frame edge $e = \{u_+, u_-\}$ the pair of selected endpoint states, as θ ranges over $[0, 1]$, sweeps a monotone staircase of (V_+, V_-)-state pairs; because the edge distribution is affine in the endpoint marginals (marginal-determination) and both readings are driven by the *single* scalar θ, the induced pair-frequency over the uniform law $\theta \sim \mathrm{Unif}[0, 1]$ equals q on that edge exactly. Coherence (pairwise marginal agreement) guarantees the per-edge sweeps share endpoint marginals at every shared vertex, so the θ-indexed selection is a *global* configuration $x(\theta)$ for each θ at which no threshold coincides with a jump, of which there are finitely many. The mixture

$$\mu \;=\; \int_0^1 \delta_{x(\theta)}\, d\theta$$

is a finite convex combination of global configurations — one per sub-interval between consecutive jumps, weighted by its length — inducing q. The construction is a finite case analysis on the ordering of the jump-thresholds followed by Lebesgue length on $[0, 1]$. □ □

A counting fact then closes the loopless symmetric case entirely.

Theorem 2.9 (forest counting)**.** *A symmetric loopless language is marginal-determined iff, as a homomorphism target, it is a forest. Every such language phase-decouples on bipartite frames.*

Proof. Write H for the target graph on vertex set A (the states) whose edge set is the loopless symmetric relation ρ; symmetry and looplessness mean $\rho = \{(a, b) : \{a, b\} \in E(H)\}$, so $|\rho| = 2|E(H)|$.

Marginal-determination ⇒ forest. Marginal-determination asks that the endpoint-marginal map — edge distribution $\mapsto$ pair of endpoint vertex-marginals — be injective. A cycle in H obstructs this: the alternating ± 1 edge-flow around it has zero vertex-marginal image, a kernel vector, so a cyclic component makes the map non-injective. Hence H is acyclic, a forest.

Forest ⇒ marginal-determined and bipartite. A forest is bipartite. On a tree the endpoint-marginal map is injective: orienting from a root, each edge distribution is recovered as the coupling of its two endpoint marginals, one edge at a time, with no cycle to obstruct. Bipartiteness makes the target phase-decoupling — a 2-colouring of H puts one endpoint of each edge in each class — so Theorem 2.8 and phase decoupling (taming (9)) apply. □ □

The assembly turns on phase decoupling, taming (9).

Lemma 2.10 (bipartite-target phase decoupling; taming (9))**.** *Let ρ be a marginal-determined language whose target graph H is bipartite, with 2-colouring $A = A_0 \sqcup A_1$. Then on every bipartite frame the coherence polytope fibers over a single* phase *scalar as*

$$C = \bigcup_{t\in[0,1]} \big(t \cdot \mathrm{FSTAB} \oplus (1-t) \cdot \mathrm{FSTAB}\big), \qquad R = \bigcup_{t\in[0,1]} \big(t \cdot \mathrm{STAB} \oplus (1-t) \cdot \mathrm{STAB}\big),$$

two copies of the edge polytope braided over t; and FSTAB = STAB *by Theorem 2.8, so $C = R$.*

Proof. A connected bipartite frame admits exactly two global colourings by H's classes, carrying masses t and $1 - t$; the phase t is the single global degree of freedom. Conditioned on either colouring, the residual data is an edge distribution over H in vertex-marginal coordinates, a point of FSTAB, and the two are independent given t — the fibered direct sum. Bipartiteness makes the edge-constraint matrix signable along the frame's bipartition, so FSTAB = STAB by Theorem 2.8 and $C = R$ termwise in t. □ □

Theorem 2.11 (Circuit Localization, symmetric loopless case)**.** *Every symmetric loopless marginal-determined language is commensurable on every gate-passing frame. Contextuality is excluded by structure: such a language is a forest target (Theorem 2.9), tamed by phase-decoupling on bipartite frames and by acyclicity (Theorem 2.1) otherwise.*

Proof. By Theorem 2.9 a symmetric loopless marginal-determined language is, as a homomorphism target, a forest, hence bipartite. Let a gate-passing frame be given. If the reporting hypergraph is acyclic, the protocol is commensurable for *every* language by Theorem 2.1, ρ included. Otherwise the frame carries a cycle; restricted to the bipartite target, the coherence polytope phase-decouples by Lemma 2.10, which reduces $C = R$ to the signable case (Theorem 2.8). Every gate-passing frame falls under one of these two cases, so the language is commensurable on all of them. The fork — a marginal-determined symmetric language contextual on some gate-passing frame — is therefore excluded by structure, not by search. □ □

Remark 2.12 (scope of the theorem)**.** The theorem is proved for loopless targets. Two extensions are verified only to finite scale, not as theorems: the *loopy* symmetric case (no pair-bipartite non-signable survivor appears at ≤ 4 states, but larger alphabets are open) and the exclusion of *cyclic* symmetric targets (established, over all lengths, only for golden-mean odd rings, Theorem 2.4). These two steps are owed.

2.4 The winding characterisation and its certified discriminant

Lift a length-L ring protocol to its *layered graph*: vertices (i, a) for layer $i \in \mathbb{Z}_L$ and state a, arcs $(i, a) \to (i+1, b)$ for legal pairs. The *winding* of a closed walk is its net layer-advance divided by L.

Theorem 2.13 (winding characterisation)**.** *Under the genericity gate (non-adjacent contexts share only trivial events), a length L is unsafe for ρ iff the layered ring over $\mathbb{Z}_L$ carries a layer-injective simple cycle of winding ≥ 2. The characterisation is structural; only its eventual-periodicity refinement (§3) is conjectural.*

The proof has three links: a gate lemma identifying C with a circulation polytope; flow decomposition reading its vertices as simple cycles; and the layer-injectivity discriminant pinning simplicity.

Lemma 2.14 (gate lemma)**.** *Under the genericity gate, the coherence polytope C of the length-L ring protocol equals the normalised circulation polytope of the layered graph: the EA (coherence) constraints and the flow-conservation constraints have the same solution set. Consequently a coherent point is a normalised circulation, and conversely.*

Proof. Give each arc $(i, a) \to (i+1, b)$ the probability $q_i(a, b)$ of the ordered pair at the adjacent contexts $(i, i+1)$. Coherence is pairwise marginal agreement: $\sum_b q_{i-1}(b, a) = \sum_b q_i(a, b)$ for every (i, a), which is Kirchhoff conservation at vertex (i, a). So a coherent point is a circulation, normalised by unit mass per context. Conversely, a circulation is coherent iff it agrees with the protocol on shared events; under the genericity gate the only shared events are the adjacent overlaps, whose agreement is the conservation already imposed, so the two systems coincide. (When the gate fails, non-adjacent contexts impose further identifications and C is a proper face of the circulation polytope.) □ □

Lemma 2.15 (flow decomposition of vertices)**.** *A vertex of the normalised circulation polytope of the layered graph is a normalised* simple *directed cycle. It is integral (hence a realisable global configuration) iff its winding is* 1*; a fractional vertex is exactly a normalised simple directed cycle of winding ≥ 2.*

Proof. By the flow-decomposition theorem every circulation is a nonnegative sum of simple-cycle flows. An extreme point admits no such sum with two terms, so a vertex is a single normalised simple cycle γ. If $w(\gamma) = 1$ then γ visits each layer once and is the graph of a configuration $x : \mathbb{Z}_L \to A$, an integral point of R. If $w(\gamma) \geq 2$ it visits some layer twice, spreading that layer's mass fractionally: non-integral. So the fractional vertices are exactly the winding-≥ 2 simple cycles, and $C \neq R$ iff one exists. □ □

A winding-≥ 2 closed walk may revisit a vertex and fail to be simple; the certificate of simplicity is layer-injectivity, each layer visited in distinct states.

Lemma 2.16 (layer-injectivity discriminant; Lean-certified)**.** *A layered-ring closed walk is a simple cycle (its vertex map is injective) iff it is layer-injective (its state map is injective on each layer fibre).*

The discriminant and the winding-1/configuration correspondence at the heart of Lemma 2.15 are machine-verified in Lean 4, zero `sorry`s, standard axioms only.[1] What Lean certifies is the combinatorial kernel — a winding-1 trace is a global configuration, a simple trace longer than L is fractional — *given* the classical flow-decomposition theorem (Ahuja–Magnanti–Orlin), which is a cited axiom, not proved. It does not certify the winding characterisation itself: the polytope-vertex step of Lemma 2.15 and the gate identification of Lemma 2.14 remain on paper.

[1] `WindingInjectivity.lean` (`simple_iff_layerInjective` and consequences) and `WindingDichotomy.lean` (`winding_one_isConfig`, `winding_ge_two_fractional`), in `formalization/QuerySystem/`.

Proof of Theorem 2.13. By Lemma 2.14 C is the circulation polytope; by Lemma 2.15 L is unsafe iff C has a fractional vertex, iff a simple cycle of winding ≥ 2 exists; by Lemma 2.16 such a cycle is exactly a layer-injective one. □ □

3 The open case and consequences

The symmetric case is settled (§2.3); any remaining contextual protocol is strongly connected and non-symmetric, the SCC-reduction lemma (monotone collapse, taming (8)) reducing commensurability to that class. Exhaustive search finds no contextual protocol there at small scale (none at ≤ 4 states, none through 9 edges at 5).

Conjecture 3.1 (universal impossibility)**.** The fork does not exist: Circuit Localization holds universally for total languages on gate-passing, time-realisable frames. Equivalently, blocking the frustrating exchange at a rich set of lengths forces one of the catalogued tamings, each of which localises the polytope.

The intended route runs through eventual periodicity of the unsafe set (a Wielandt-bounded union of arithmetic progressions), which reduces to a combinatorial pruning lemma.

Conjecture 3.2 (pruning lemma)**.** Let TR_k be the lengths admitting a diagonal-avoiding tuple-rotation walk in the k-fold tensor power, and LISC_k the lengths admitting a layer-injective simple cycle of winding k. Then $\mathrm{TR}_k = \mathrm{LISC}_k$ cofinitely.

The inclusion $\mathrm{LISC}_k \subseteq \mathrm{TR}_k$ is immediate. The reverse, at $k = 2$, reduces to whether two exchange-linked winding-1 strands form a single simple cycle — but the reduction drops a relative-phase degree of freedom (a winding-2 simple cycle interleaves its wraps at a phase a lock-step exchange cannot express), so the $k = 2$ residual is phase-parametrised strand-disjointness. No $\mathrm{TR}_k \setminus \mathrm{LISC}_k$ disagreement appears to layered length 35: evidence, not proof.

3.1 Consequences for reconstruction

On the tame classes — acyclic reporting (Theorem 2.1), symmetric marginal-determined languages (Theorem 2.11), even-cyclic golden-mean recurrence (Theorem 2.4) — coherent window-statistics determine a global process, and Theorem 2.13 makes the boundary decidable: a layer-injective winding-≥ 2 cycle separates the contextual protocols from the safe ones. This is the licensing layer beneath process reconstruction ("statistical Takens": a measure-plus-shift, not a manifold). By Theorem 2.3 the boundary is intrinsic — general variety-typed protocols are as hard as rational polytope membership — so contextuality is a feature of observing through windows, present exactly when the geometry admits the winding obstruction; where Conjecture 3.1 holds, no total language manufactures it unless the exchange criterion fires.

The same failure of local coherence to globalise appears in Paper II algebraically (dispersion-free states collapsing PR on non-distributive logics) and in the σ-essential witness set-theoretically (a finitely-coherent pattern with no σ-additive global state). Dynamically it is sharpest: frustration is an unwitnessed exchange — two histories trading places around the clock of observation without meeting — and contextual data is the shadow of an unresolved covering of time.

Note on method

Each commensurability verdict is a theorem or an exact rational enumeration, each contextuality verdict an exhibited rational witness in $C \setminus R$; sampling serves only as screening. Lemma 2.16 and the flow-decomposition kernel are machine-checked in Lean 4; the constructive tamings (θ-sweep, phase-mixture, monotone collapse) are the remaining certification targets.

References

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